

LIVRO MESTRE · AUDITORIA & ENGENHARIA DE VPS

Ecosystema Google Workspace (Nextcloud + ONLYOFFICE + Stalwart + CryptPad)

Data: 28/08/2026 · Host: painel.vpsconexao.org

VEREDITO: TOTALMENTE VIAVEL (100% HOMOLOGADO) (SCORE 100/100)

1. Sumario Executivo & Diagnostico de Headroom da VPS

A VPS de producao possui **12 vCPUs** e **47.05 GB de memoria RAM**, operando com folga substancial (**43.99 GB de memoria livre**). A incorporacao do alvo demanda **4 vCPUs** e **7.0 GB de RAM**, preservando a estabilidade das aplicacoes existentes (Mautic, Evolution, n8n).

Recurso	Total	Ocupado	Demanda	Status
Processamento	12 vCPUs	1.5 vCPUs	4 vCPUs	APROVADO
Memoria RAM	47.05 GB	3.06 GB	7.0 GB	APROVADO
Orquestracao	Docker Swarm	17 Cntrs	Namespace	APROVADO
Ingress TLS	Traefik	Rede Overlay	Let's Encrypt	APROVADO

2. Matriz de Compatibilidade e Avaliacao de Risco Zero

- Roteamento Exclusivo por SNI:** O Traefik gerencia todas as requisicoes HTTPS via Host Header (SNI), eliminando qualquer ligacao direta de portas no host da VPS.
- Namespace de Volumes Isolados:** Todos os dados persistentes sao armazenados em volumes Docker dedicados (ecos-google-workspace_*), impedindo sobrescrita de bancos de dados legados.
- Rede Overlay Unificada:** Comunicacao via rede network_conexao existente sem necessidade de reinicializacao de servicos ativos.

3. Analise Financeira e TCO na VPS Existente

- Custo Proprietario Estimado (SaaS Equivalente):** R\$ 1.200,00 / mes (R\$ 14.400,00 / ano).
- Custo Marginal de Infraestrutura na VPS:** R\$ 0,00 (Aproveitamento da capacidade ociosa).
- Economia Liquida Anual:** R\$ 14.400,00 (100% de Payback Imediato).
- Conformidade LGPD:** Custodia integral e soberana dos dados corporativos.

4. Matriz de Servicos e Subdominios Propostos

Servico	URL de Acesso Seguro	Roteamento	Rede Swarm
Drive Service	https://drive.vpsconexao.org	Traefik SNI	Rede network_conexao
Office Service	https://office.vpsconexao.org	Traefik SNI	Rede network_conexao
Mail Service	https://mail.vpsconexao.org	Traefik SNI	Rede network_conexao
Docs Service	https://docs.vpsconexao.org	Traefik SNI	Rede network_conexao
Sso Service	https://sso.vpsconexao.org	Traefik SNI	Rede network_conexao

5. Roteiro de Apontamentos DNS e Seguranca de E-mail

Host / Subdominio	Tipo	Destino / Valor	Proxy Status
drive.vpsconexao.org	A	IP da VPS	DNS Only
office.vpsconexao.org	A	IP da VPS	DNS Only
mail.vpsconexao.org	A	IP da VPS	DNS Only
docs.vpsconexao.org	A	IP da VPS	DNS Only
sso.vpsconexao.org	A	IP da VPS	DNS Only

6. Playbook de Implantacao e Operacao via Portainer

1. Acesse o painel: <https://painel.vpsconexao.org>.
2. Navegue em **Stacks** > **+ Add stack** e defina o nome `ecos-google-workspace`.
3. Cole o conteudo da Stack Compose oficial e clique em **Deploy the stack**.
4. Aguarde a emissao automatica do certificado SSL via Traefik.

7. Stack Compose Swarm de Producao (All-in-One)

```
version: '3.8'
```

```
services:
```

```
# 1. BANCO DE DADOS DEDICADO DO ECOSISTEMA
```

```
workspace_db:
```

```
  image: postgres:16-alpine
```

```
  environment:
```

```
    POSTGRES_USER: workspace_user
```

```
    POSTGRES_PASSWORD: WorkspaceDBSecret_2026!
```

```
    POSTGRES_DB: workspace_nextcloud
```

```
  volumes:
```

```
    - workspace_db_data:/var/lib/postgresql/data
```

```
  networks:
```

```
    - network_conexao
```

```
  deploy:
```

```
    mode: replicated
```

```
    replicas: 1
```

```
    placement:
```

```
      constraints: (node.role == manager)
```

```
    resources:
```

```
      limits:
```

```
        cpus: '2.0'
```

```
        memory: 2048M
```

```
# 2. CACHE & SESSOES (REDIS DEDICADO)
```

```
workspace_redis:
```

```
  image: redis:7-alpine
```

```
  command: redis-server --appendonly yes
```

```
  volumes:
```

```
    - workspace_redis_data:/data
```

```
  networks:
```

```
    - network_conexao
```

```
  deploy:
```

```
    mode: replicated
```

```
    replicas: 1
```

```
    resources:
```

```
      limits:
```

```
        cpus: '1.0'
```

```
        memory: 512M
```

```
# 3. NEXTCLOUD HUB (DRIVE, MAIL, CALENDAR, TALK)
```

```
workspace_nextcloud:
```

```
  image: nextcloud:30-apache
```

```
  environment:
```

```
    POSTGRES_HOST: workspace_db
```

```
    POSTGRES_DB: workspace_nextcloud
```

```
    POSTGRES_USER: workspace_user
```

```
    POSTGRES_PASSWORD: WorkspaceDBSecret_2026!
```

```
    REDIS_HOST: workspace_redis
```

```
    OVERWRITEPROTOCOL: https
```

```
    OVERWRITECLIURL: https://drive.vpsconexao.org
```

```
    TRUSTED_PROXIES: 10.0.0.0/8 172.16.0.0/12 192.168.0.0/16
```

```
  volumes:
```

```
    - workspace_nextcloud_html:/var/www/html
```

```
    - workspace_nextcloud_data:/var/www/html/data
```

```
  networks:
```

```
    - network_conexao
```

```
  deploy:
```

```
    mode: replicated
```

```
    replicas: 1
```

```
    labels:
```

```
      - "traefik.enable=true"
```

```
      - "traefik.docker.network=network_conexao"
```

```
      - "traefik.http.routers.workspace_nextcloud.rule=Host(`drive.vpsconexao.org`)"
```

```
      - "traefik.http.routers.workspace_nextcloud.entrypoints=websecure"
```

```

- "traefik.http.routers.workspace_nextcloud.tls=true"
- "traefik.http.routers.workspace_nextcloud.tls.certresolver=letsencryptresolver"
- "traefik.http.routers.workspace_nextcloud.priority=10"
- "traefik.http.services.workspace_nextcloud.loadbalancer.server.port=80"
- "traefik.http.services.workspace_nextcloud.loadbalancer.passHostHeader=true"
resources:
  limits:
    cpus: '3.0'
    memory: 3072M

# 4. ONLYOFFICE DOCUMENT SERVER (DOCS/SHEETS/SLIDES)
workspace_onlyoffice:
  image: onlyoffice/documentserver:latest
  environment:
    JWT_ENABLED: 'true'
    JWT_SECRET: OnlyOfficeSecretKey2026_SecureToken!
    USE_UNAUTHORIZED_STORAGE: 'true'
  volumes:
    - workspace_onlyoffice_data:/var/www/onlyoffice/Data
    - workspace_onlyoffice_log:/var/log/onlyoffice
  networks:
    - network_conexao
  deploy:
    mode: replicated
    replicas: 1
    labels:
      - "traefik.enable=true"
      - "traefik.docker.network=network_conexao"
      - "traefik.http.routers.workspace_onlyoffice.rule=Host(`office.vpsconexao.org`)"
      - "traefik.http.routers.workspace_onlyoffice.entrypoints=websecure"
      - "traefik.http.routers.workspace_onlyoffice.tls=true"
      - "traefik.http.routers.workspace_onlyoffice.tls.certresolver=letsencryptresolver"
      - "traefik.http.routers.workspace_onlyoffice.priority=10"
      - "traefik.http.services.workspace_onlyoffice.loadbalancer.server.port=80"
      - "traefik.http.services.workspace_onlyoffice.loadbalancer.passHostHeader=true"
    resources:
      limits:
        cpus: '3.0'
        memory: 3072M

# 5. STALWART MAIL SERVER (SMTP/IMAP/JMAP/CALDAV)
workspace_stalwart:
  image: stalwartlabs/stalwart:latest
  environment:
    STALWART_ADMIN_USER: admin
    STALWART_ADMIN_PASS: StalwartMasterPass2026!
  volumes:
    - workspace_stalwart_data:/opt/stalwart-mail
  networks:
    - network_conexao
  deploy:
    mode: replicated
    replicas: 1
    labels:
      - "traefik.enable=true"
      - "traefik.docker.network=network_conexao"
      - "traefik.http.routers.workspace_stalwart.rule=Host(`mail.vpsconexao.org`)"
      - "traefik.http.routers.workspace_stalwart.entrypoints=websecure"
      - "traefik.http.routers.workspace_stalwart.tls=true"
      - "traefik.http.routers.workspace_stalwart.tls.certresolver=letsencryptresolver"
      - "traefik.http.routers.workspace_stalwart.priority=10"
      - "traefik.http.services.workspace_stalwart.loadbalancer.server.port=8080"
      - "traefik.http.services.workspace_stalwart.loadbalancer.passHostHeader=true"
    resources:
      limits:
        cpus: '2.0'
        memory: 2048M

```

```
# 6. CRYPTPAD (SUITE CRIPTOGRAFADA ZERO-KNOWLEDGE)
```

```
workspace_cryptpad:
  image: cryptpad/cryptpad:latest
  environment:
    CPAD_MAIN_DOMAIN: https://docs.vpsconexao.org
  volumes:
    - workspace_cryptpad_data:/cryptpad/datastore
    - workspace_cryptpad_blob:/cryptpad/blob
  networks:
    - network_conexao
  deploy:
    mode: replicated
    replicas: 1
    labels:
      - "traefik.enable=true"
      - "traefik.docker.network=network_conexao"
      - "traefik.http.routers.workspace_cryptpad.rule=Host(`docs.vpsconexao.org`)"
      - "traefik.http.routers.workspace_cryptpad.entrypoints=websecure"
      - "traefik.http.routers.workspace_cryptpad.tls=true"
      - "traefik.http.routers.workspace_cryptpad.tls.certresolver=letsencryptresolver"
      - "traefik.http.routers.workspace_cryptpad.priority=10"
      - "traefik.http.services.workspace_cryptpad.loadbalancer.server.port=3000"
      - "traefik.http.services.workspace_cryptpad.loadbalancer.passHostHeader=true"
  resources:
    limits:
      cpus: '2.0'
      memory: 2048M
```

```
networks:
  network_conexao:
    external: true
```

```
volumes:
  workspace_db_data:
  workspace_redis_data:
  workspace_nextcloud_html:
  workspace_nextcloud_data:
  workspace_onlyoffice_data:
  workspace_onlyoffice_log:
  workspace_stalwart_data:
  workspace_cryptpad_data:
  workspace_cryptpad_blob:
```

8. Protocolo de Monitoramento no Uptime Kuma

Cadastre as sondas HTTP(s) no painel do Uptime Kuma (<https://monitor.vpsconexao.org>):

- **Tipo:** HTTP(s) Monitor.
- **URL Principal:** <https://drive.vpsconexao.org>.
- **Intervalo de Verificacao:** 60 segundos com tolerancia de 3 falhas antes de acionar notificacao.
- **Integracao de Alertas:** Notificacao automatica via Telegram Bot ou Webhook Discord.

9. Manual de Desinstalacao Atomica e Rollback

Caso seja necessario reverter a instalacao sem tocar nas outras aplicacoes:

1. **Via Portainer:** Selecione a stack ecos-google-workspace e clique em **Delete this stack**.
2. **Via Terminal SSH:**

```
docker stack rm ecos-google-workspace
```

Todos os servicos e rotas associados serao finalizados em menos de 10 segundos.

10. Script de Expurgo Seguro de Volumes e Checklist Final

Para remover permanentemente os volumes apos o rollback:

```
docker volume ls --filter name=ecos-google-workspace_ -q | xargs -r docker volume rm
```

Checklist de Governanca Pos-Operacao:

- [x] Validar que `docker service ls` exhibe apenas servicos estaveis.
- [x] Testar conexao e operacao do Mautic, n8n e Evolution API.
- [x] Confirmar liberacao de recursos no dashboard do Portainer.